

Dissolved Oxygen and Aquatic Invertebrate Abundance at North Campus Open Space

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Introduction

Wetland restoration projects are often evaluated using physical, chemical, and biological indicators because restoration success depends not only on habitat construction, but also on whether restored ecosystems support functioning biological communities. Aquatic invertebrates are useful indicators of wetland condition because their abundance and community composition can respond to changes in water quality, hydrology, vegetation, and habitat structure. Studies from multiple wetland systems show that restored or rehabilitated wetlands can support greater invertebrate abundance than imbalanced wetlands (Batzer et al. 2015); (Lu et al. 2020). This suggests that aquatic invertebrate communities can provide insight into restoration progress.

As a result, aquatic invertebrate monitoring especially relevant at North Campus Open Space (NCOS) because it is a restored wetland where long-term monitoring can help evaluate how biological communities respond to changing environmental conditions.

Dissolved oxygen (DO) is one important water-quality variable because aquatic organisms depend on oxygen for survival, metabolism, reproduction, and movement. Research has shown that low oxygen conditions can negatively affect aquatic invertebrates by reducing survival, emergence, growth, feeding, reproduction, and activity (Connolly, Crossland, and Pearson 2004); (Galic, Hawkins, and Forbes 2019); (Collins et al. 2019). However, the effects of dissolved oxygen are not always simple or uniform across taxa. Some invertebrates can tolerate moderate reductions in oxygen, while others are more sensitive to hypoxia or respond behaviorally by moving away from low-oxygen areas (Connolly, Crossland, and Pearson 2004); (Ekau et al. 2010). One study found that oxygen can be a strong predictor of macroinvertebrate richness, but other studies show that dissolved oxygen also interacts with factors such as temperature, water depth, precipitation, and habitat zone (Croijmans, De Jong, and Prins 2021); (Lucchesi, Chipps, and Schumann 2022). Therefore, dissolved oxygen may influence both the total abundance of aquatic invertebrates and the relative abundance of different taxa within the community.

Research conducted at other locations also suggests that dissolved oxygen can shape aquatic invertebrate community structure, not just total abundance. In Lake Huron coastal wetland wet meadows, dissolved oxygen gradients influenced oxygen-sensitive zoobenthos, although broader site and wetland differences also helped explain community patterns (Burton and Uzarski 2009). Similarly, one study found that seasonal hypoxia affected where macroinvertebrates were found in a reservoir, but taxa such as Cladocera and Copepoda were also influenced by water depth and monthly precipitation (Lucchesi, Chipps, and Schumann 2022). These findings suggest that, while dissolved oxygen is ecologically important, its relationship with invertebrate abundance may be impacted by site conditions, seasonal change, and taxon-specific tolerance.

At NCOS, previous analysis by UCSB students also examined relationships among algae, macroinvertebrates, and water quality. Their poster suggested that dissolved oxygen change over time and may be linked to macroinvertebrate abundance (Huynh, Russell, and Malone 2023). They also found that ostracods and copepods were among the most abundant taxa. However, their findings also indicated that other water-quality variables, such as conductivity, may influence diversity and abundance patterns. This is important because it indicates that dissolved oxygen levels should be interpreted as part of a broader set of environmental factors rather than as the only driver of invertebrate abundance.

Building on previous previous, our analysis examines the relationship between dissolved oxygen and aquatic invertebrate abundance at NCOS. Specifically, we ask: **What is the strength and direction of the relationship between dissolved oxygen and invertebrate abundance?** To answer this question, we analyzed median dissolved oxygen levels (mg/L) and invertebrate abundance for our five focus taxa: Ostracod, Copepod, Cladocera, Hemiptera Corixidae Boatman, and Nematode. Throughout our analysis, we examined the relationship

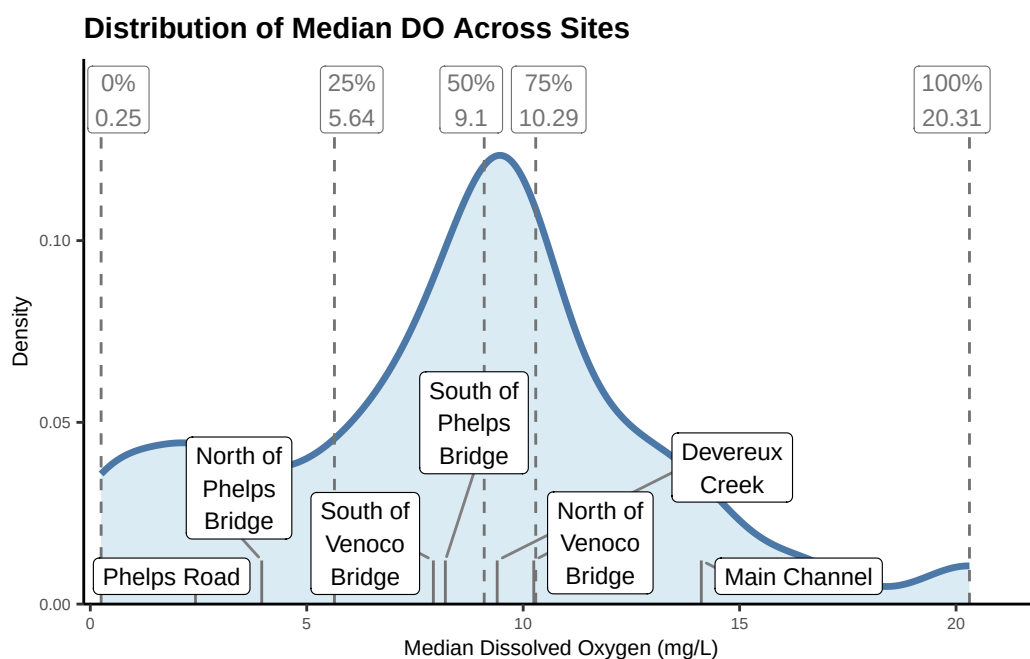
between DO and total invertebrate abundance, how DO varies across NCOS sites, taxon-specific abundance, abundance across DO categories, and community composition across low, medium, and high DO conditions.

Methods

Results

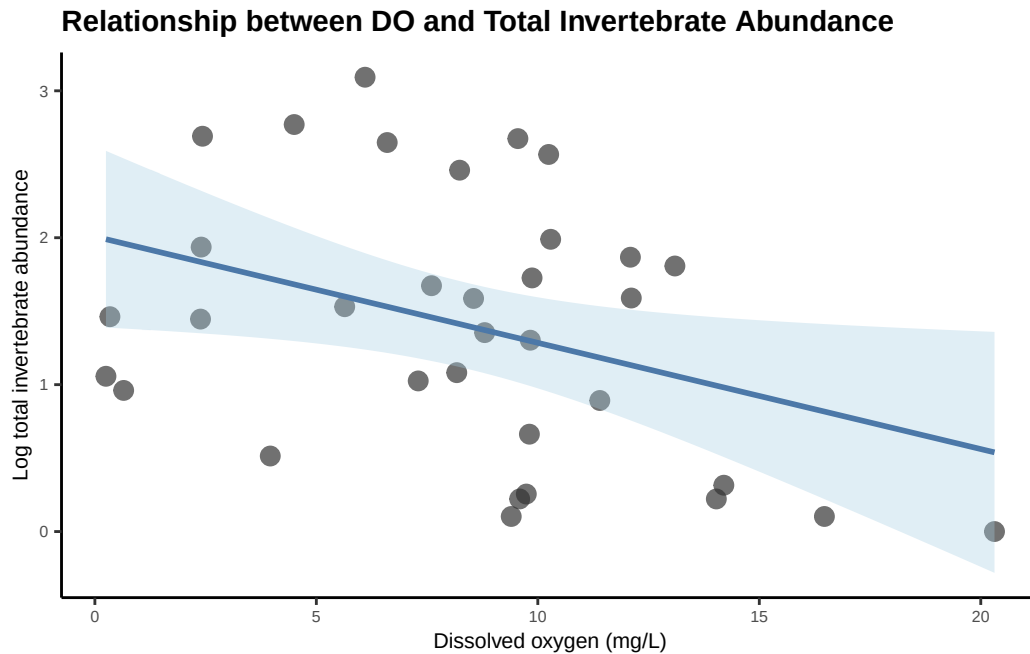
We first examined dissolved oxygen patterns across NCOS sampling sites to understand the oxygen conditions experienced by aquatic invertebrates. Median dissolved oxygen varied substantially among sites, indicating that invertebrates were not sampled under uniform oxygen conditions across the study area (Figure 1). Most median DO values were concentrated around approximately 8–11 mg/L, but the full range extended from very low values to values above 20 mg/L. Phelps Road and North of Phelps Bridge had relatively low median DO values, while Main Channel had the highest median DO values. This site-level variation provides important context for the main research question because it shows that the dataset includes a clear gradient in dissolved oxygen conditions.

Figure 1. Distribution of Median DO Across Sites



After establishing this DO gradient, we examined whether total invertebrate abundance was related to continuous dissolved oxygen values. Total abundance, summed across the five focus taxa and log-transformed, showed a weak-to-moderate negative relationship with median dissolved oxygen (Figure 2).

Figure 2. Relationship between DO and Total Invertebrate Abundance



A Pearson correlation test showed a statistically significant negative linear relationship between median DO and log-transformed total invertebrate abundance, $r = -0.375$, $t(32) = -2.29$, $p = 0.0289$. The 95% confidence interval for the correlation ranged from -0.633 to -0.042, which does not include zero, further supporting a negative relationship. This means that as dissolved oxygen increased, total invertebrate abundance tended to decrease. However, because the correlation was not close to -1, the relationship was not strong; dissolved oxygen explained some, but not all, of the variation in total abundance.

Pearson's product-moment correlation

```
data: total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

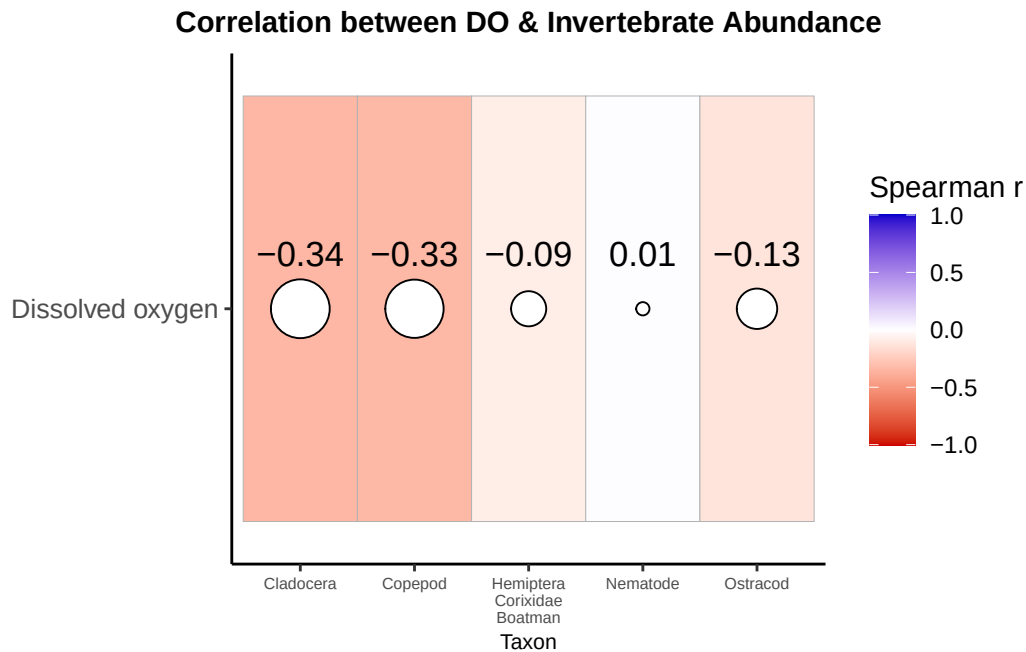
```

-0.63289820 -0.04217141
sample estimates:
      cor
-0.3749894

```

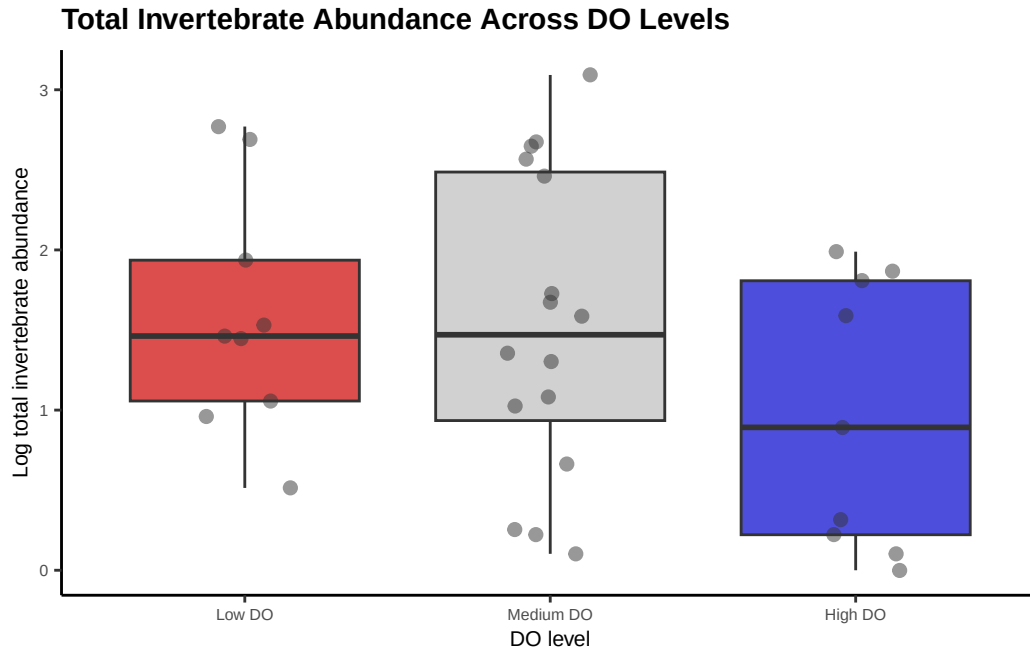
We then examined whether this relationship was consistent across individual taxa. Taxon-specific correlations showed that most focus taxa had weak negative relationships with dissolved oxygen, but the strength of the relationship differed among groups (Figure 3). Cladocera and Copepod had the strongest negative correlations with DO, while Ostracod and Hemiptera Corixidae Boatman had weaker negative correlations. Nematode showed almost no relationship with dissolved oxygen. This suggests that the community-level relationship between DO and abundance was not driven equally by all taxa, but instead reflected different taxon-specific patterns.

Figure 3: Correlation between DO & Invertebrate Abundance



To compare abundance across broader oxygen conditions, we grouped observations into low, medium, and high dissolved oxygen categories. Total log-transformed invertebrate abundance appeared lower in the High DO category than in the Low DO and Medium DO categories (Figure 4). However, there was substantial overlap among the categories, especially in the Medium DO group.

Figure 4: Total Invertebrate Abundance Across DO Levels



A Kruskal-Wallis test showed that total abundance did not differ significantly across the three DO categories, $X^2 = 2.00$, $df = 2$, $p = 0.368$. This indicates that, although the visualization suggests lower abundance under high DO conditions, there was not enough statistical evidence to conclude that total abundance differed significantly among the DO categories.

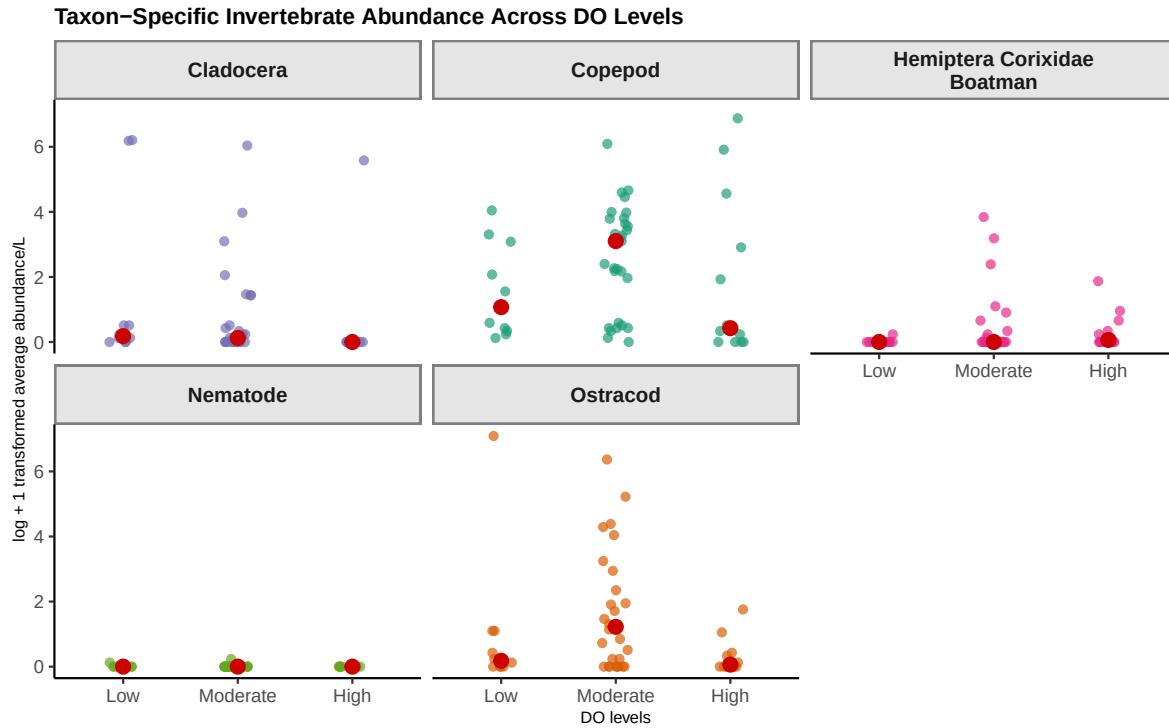
Kruskal-Wallis rank sum test

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data: log_total_abundance by do_category
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Kruskal-Wallis chi-squared = 2.0001, df = 2, p-value = 0.3679
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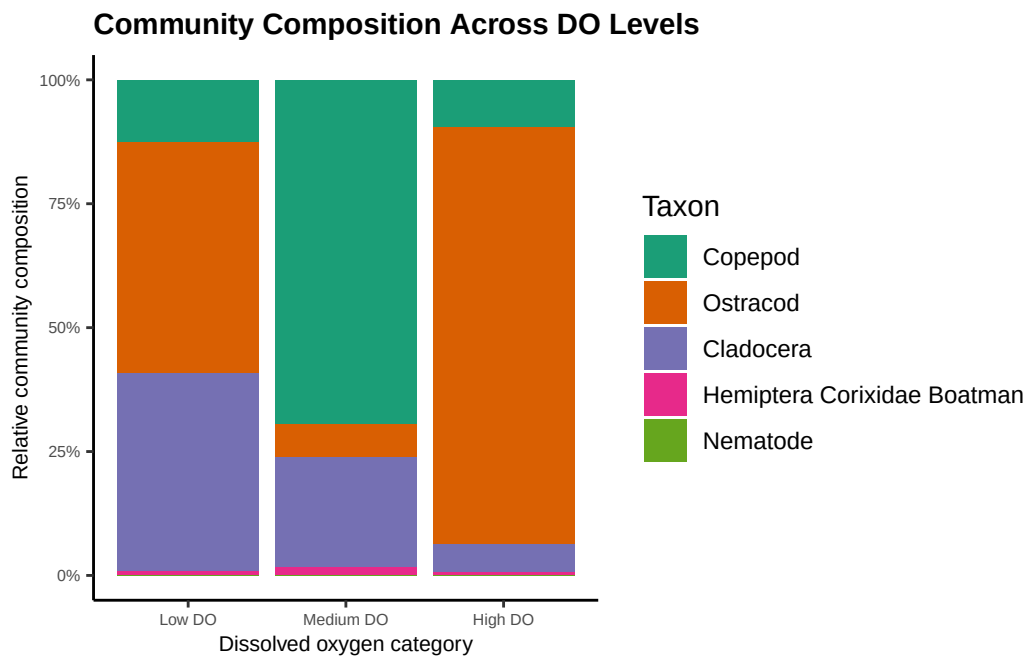
Taxon-specific abundance patterns across DO categories showed more detail than the community-level comparison. Copepods and Ostracods had higher and more variable abundances in the Moderate DO category, while Cladocera showed some high-abundance observations in the Low and Moderate DO categories (Figure 5). Nematodes remained close to zero across all dissolved oxygen levels, and Hemiptera Corixidae Boatman generally had low abundance, with only a few higher observations in Moderate DO. These patterns suggest that dissolved oxygen may be related to abundance differently depending on the taxon, rather than producing one consistent response across the whole invertebrate community.

Figure 5. Taxon-Specific Invertebrate Abundance Across DO Levels



Finally, we examined whether the relative composition of the invertebrate community changed across dissolved oxygen categories. Community composition shifted clearly among Low, Medium, and High DO groups (Figure 6). Low DO sites had a more mixed community dominated mostly by Ostracods and Cladocera. Moderate DO sites were dominated by Copepods and Ostracods, with smaller contributions from Cladocera and Hemiptera Corixidae Boatman. High DO sites were dominated primarily by Copepods, with Cladocera making up a smaller portion of the community and Ostracods contributing very little. Hemiptera Corixidae Boatman and Nematodes made up small proportions of the community across all DO categories.

Figure 6. Community Composition Across DO Levels



A Pearson's chi-squared test showed that taxon composition differed significantly across DO bins, $X^2 = 2859.2$, $df = 8$, $p < 2.2e-16$. This provides strong evidence that the relative abundances of the focus taxa were not evenly distributed across low, medium, and high DO conditions. In other words, dissolved oxygen categories were strongly associated with changes in community composition, even though total abundance did not differ significantly across DO categories.

	Cladocera	Copepod	Hemiptera Corixidae Boatman	Nematode	Ostracod
low	980.2667	115.8667	0.2666667	0.1333333	1202.933333
moderate	510.6667	1170.4000	83.8666667	0.2666667	1057.866667
high	264.2667	1453.4667	8.8000000	0.0000000	7.866667

Pearson's Chi-squared test

data: chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16

Discussion

This study found that the relationship between dissolved oxygen and aquatic invertebrate abundance at NCOS was generally negative but weak-to-moderate in strength. When dissolved oxygen levels were analyzed as a continuous variable, total log-transformed invertebrate abundance decreased as median dissolved oxygen increased. The Pearson correlation was statistically significant, suggesting evidence of a negative linear relationship between dissolved and total abundance. However, the relationship was not strong. This indicates that, while dissolved oxygen is related to invertebrate abundance, it does not fully explain the variation observed in the dataset.

These findings only partly align with previous work on DO and aquatic invertebrates. Many studies suggest that low dissolved oxygen or hypoxia can negatively affect aquatic invertebrates by reducing survival, reproduction, feeding, growth, emergence, and movement (Connolly, Crossland, and Pearson 2004); (Galic, Hawkins, and Forbes 2019); (Collins et al. 2019); (Ekau et al. 2010). Based on this literature, we might expect higher invertebrate abundance under higher oxygen conditions. However, our results have shown the opposite pattern for total abundance, with higher abundance often occurring under lower or moderate dissolved oxygen levels. This difference does not necessarily mean that dissolved is unimportant at NCOS. Instead, it suggests that the relationship between dissolved oxygen and invertebrate abundance is highly complex and may be influenced by other site-level and ecological factors.

The poster about NCOS presented by UCSB students, provides useful context for interpreting this pattern. Their analysis suggested that macroinvertebrate abundance may increase with higher dissolved oxygen, but they also noted that conductivity and short-term population spikes may influence species diversity and abundance (Huynh, Russell, and Malone 2023). Our results did not directly support the pattern of higher abundance at higher dissolved oxygen. Instead, we found that total abundance decreased as dissolved oxygen increased, while community composition changed strongly across dissolved oxygen categories. This suggests that dissolved oxygen may be more closely related to which taxa dominate the community than to total abundance alone. For example, low dissolved oxygen levels had relatively large contributions from ostracods and cladocerans, while high dissolved oxygen levels were dominated by copepods. This finding is also consistent with the idea that different taxa vary in their tolerance to oxygen conditions (Connolly, Crossland, and Pearson 2004); (Ekau et al. 2010); (Ekau et al. 2010).

The significant result of the chi-square test provides the strongest evidence that community structure differed across dissolved oxygen categories. Although total abundance did not differ significantly among low, medium, and high dissolved oxygen levels in the Kruskal-Wallis test, taxon composition did differ significantly. This means that dissolved oxygen categories were associated with changes in the relative abundance of different taxa, even when total abundance differences were not statistically significant. This pattern is similar to findings from other wetland and reservoir systems, where dissolved oxygen influenced invertebrate distribution and community composition but interacted with other environmental conditions

such as water depth, precipitation, habitat zone, and site-level differences (Burton and Uzarski 2009); (Lucchesi, Chipps, and Schumann 2022).

Because NCOS is a restored wetland, these results also connect to broader questions about restoration monitoring. Previous wetland restoration studies have found that aquatic invertebrate abundance and assemblages can be useful indicators of restoration condition, with restored or rehabilitated wetlands often supporting more abundant invertebrate communities than impaired wetlands (Batzner et al. 2015); (Lu et al. 2020); (Raison 2023). Our analysis supports the value of aquatic invertebrate monitoring at NCOS because the invertebrate community varied across environmental conditions and showed clear shifts in composition. However, our results also show that interpreting restoration success from abundance alone can be difficult. Higher total abundance does not automatically indicate better habitat conditions if the community is dominated by a few tolerant or opportunistic taxa.

Several challenges make this type of dataset difficult to collect and interpret. Aquatic invertebrate sampling requires field collection, sorting, identification, and counting of small organisms, which can be time-consuming and sensitive to sampling method. Some taxa may be difficult to identify consistently, especially when individuals are small, damaged, or present in early life stages. Abundance estimates can also be affected by short-term population pulses, patchy distributions, and differences in sampling location or timing. Because invertebrates are not evenly distributed throughout wetland habitats, a sample from one site or date may not fully represent the broader community.

Other variables besides dissolved oxygen likely influenced the patterns observed in this study. Conductivity and salinity may affect which taxa can tolerate particular sites, and Huynh et al. suggested that high conductivity at one NCOS site may have contributed to increased species diversity (Huynh, Russell, and Malone 2023). Water temperature is also important because temperature affects both oxygen availability and invertebrate metabolism. Additionally, previous research shows that higher temperatures can increase the oxygen threshold needed for some organisms to survive (Galic, Hawkins, and Forbes 2019). Water depth and precipitation may also influence abundance patterns by changing habitat availability, dilution, flow, and the duration of hypoxic conditions (Lucchesi, Chipps, and Schumann 2022). Vegetation and plant richness are also relevant because plants can affect habitat structure, food availability, refuge from predators, and daytime oxygen production (Raison 2023). Future analyses could incorporate these variables to determine whether dissolved oxygen levels remain important after accounting for broader site and seasonal conditions.

Temporal variation is another important limitation. Invertebrate abundance can change over short periods because of life cycles, emergence, reproduction, dispersal, rainfall events, temperature shifts, and seasonal changes in water quality. Huynh et al. (2023) noted that macroinvertebrate populations at NCOS showed short-term spikes followed by declines, which suggests that timing strongly affects observed abundance (Huynh, Russell, and Malone 2023). Our dataset captures repeated sampling across dates, but quarterly sampling may miss short-term events such as sudden hypoxia, storm-driven changes, algal blooms, or rapid population increases. As a result, the measured dissolved oxygen levels for a sampling date may not fully

represent the oxygen conditions experienced by invertebrates in the days or weeks prior to collection.

Overall, our findings suggest that dissolved oxygen is related to aquatic invertebrate patterns at NCOS, but the relationship is not simple. Total abundance showed a weak-to-moderate negative relationship with dissolved oxygen levels, while community composition changed strongly across dissolved oxygen categories. This indicates that dissolved oxygen may be more important for shaping which taxa dominate the community than for predicting total abundance alone. Future work should examine dissolved oxygen levels alongside other variables like conductivity, salinity, temperature, water depth, precipitation, vegetation, and seasonal timing to better understand the environmental factors structuring aquatic invertebrate communities at NCOS.

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